THEOREM 1 Squaring Operation on Equations

If both sides of an equation are squared, then the solution set of the original equation is a subset of the solution set of the new equation.

Equation	Solution Set
$x = 3$	$\{3\}$
$x^2 = 9$	$\{-3, 3\}$

This theorem provides us with a method of solving some equations involving radicals. It is important to remember that any new equation obtained by raising both sides of an equation to the same power may have solutions, called **extraneous solutions,** that are not solutions of the original equation. Fortunately though, any solution of the original equation must be among those of the new equation.

When raising both sides of an equation to a power, checking solutions is not just a good idea—it is essential to identify any extraneous solutions.

EXPLORE-DISCUSS 1

Squaring both sides of the equations $x = \sqrt{x}$ and $x = -\sqrt{x}$ produces the new equation $x^2 = x$. Find the solutions to the new equation and then check for extraneous solutions in each of the original equations.

EXAMPLE 1 Solving Equations Involving Radicals

Solve:

(A) $x + \sqrt{x - 4} = 4$ (B) $\sqrt{2x + 3} - \sqrt{x - 2} = 2$

SOLUTIONS (A)

$$x + \sqrt{x-4} = 4 \qquad \text{Isolate radical on one side.}$$

$$\sqrt{x-4} = 4 - x \qquad \text{Square both sides.}$$

$$\boxed{(\sqrt{x-4})^2 = (4-x)^2} \qquad \text{See the upcoming caution on squaring the right side.}$$

$$x - 4 = 16 - 8x + x^2 \qquad \text{Write in standard form.}$$

$$x^2 - 9x + 20 = 0 \qquad \text{Factor left side.}$$

$$(x-5)(x-4) = 0 \qquad \text{Use the zero product property.}$$

$$\boxed{x - 5 = 0 \quad \text{or} \quad x - 4 = 0}$$

$$x = 5 \quad \text{or} \quad x = 4$$

CHECK

$x = 5$	$x = 4$
$x + \sqrt{x-4} = 4$	$x + \sqrt{x-4} = 4$
$5 + \sqrt{5-4} \stackrel{?}{=} 4$	$4 + \sqrt{4-4} \stackrel{?}{=} 4$
$6 \neq 4$	$4 \stackrel{\checkmark}{=} 4$

This shows that 4 is a solution to the original equation and 5 is extraneous. The only solution is $x = 4$.

CHAPTER 1

››› GROUP ACTIVITY Solving a Cubic Equation

If a, b, and c are real numbers with $a \neq 0$, then the quadratic equation $ax^2 + bx + c = 0$ can be solved by a variety of methods, including the quadratic formula. How can we solve the cubic equation

$$ax^3 + bx^2 + cx + d = 0, \qquad a \neq 0 \tag{1}$$

and is there a formula for the roots of this equation?

The first published solution of equation (1) is generally attributed to the Italian mathematician Girolamo Cardano (1501–1576) in 1545. His work led to a complicated formula for the roots of equation (1) that involves topics that are discussed later in this text. For now, we will use Cardano's method to find a real solution in special cases of equation (1). Note that because a is nonzero, we can always multiply both sides of (1) by $1/a$ to make the coefficient of x^3 equal to 1.

CARDANO'S METHOD FOR SOLVING A CUBIC EQUATION

Let $x^3 + bx^2 + cx + d = 0$

Example problem: $x^3 - 6x^2 + 6x - 5 = 0$. Steps will be in red.

Step 1. Substitute $x = y - b/3$ to obtain the *reduced cubic* $y^3 + my = n$.

$x = y - \frac{-6}{3}$ or $x = y + 2$. The equation becomes

$$(y + 2)^3 - 6(y + 2) + 6(y + 2) - 5 = 0,$$

which simplifies to $y^3 - 6y = 9$: $m = -6$, $n = 9$.

Step 2. Define u and v by $m = 3uv$ and $n = u^3 - v^3$. Use $v = \frac{m}{3u}$ to write

$$n = u^3 - \left(\frac{m}{3u}\right)^3$$

Multiply both sides by u^3 to obtain an equation quadratic in u^3. Solve for u^3 by factoring or by using the quadratic formula. Then solve for u, and find the associated value of v.

$v = \frac{-6}{3u}$ or $v = -\frac{2}{u}$; $9 = u^3 - \left(-\frac{2}{u}\right)^3 = u^3 + \frac{8}{u^3}$ Multiply both sides by u^3 to obtain $u^6 - 9u^3 + 8 = 0$; solve by factoring to get $u = 2$ (in which case $v = -1$) or $u = 1$ (in which case $v = -2$).

Step 3. Using either of the solutions found in step 2,

$$x = y - \frac{b}{3} = u - v - \frac{b}{3}$$

is a solution to $x^3 + bx^2 + cx + d = 0$

For $u = 2$, $v = -1$, $x = 2 - (-1) - \frac{-6}{3} = 5$ (Solution)

(A) The key to Cardano's method is to recognize that if u and v are defined as in step 2, then $y = u - v$ is a solution of the reduced cubic. Verify this by substituting $y = u - v$, $m = 3uv$, and $n = u^3 - y^3$ in $y^3 + my = n$ and show that the result is an identity.

(B) Use Cardano's method to solve

$$x^3 - 6x^2 - 3x - 8 = 0 \tag{2}$$

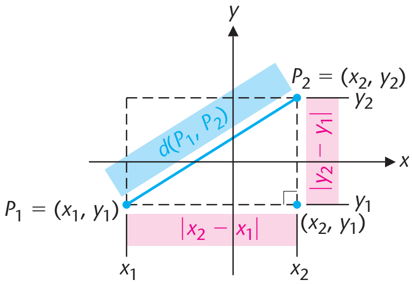
Use a calculator to find a decimal approximation of your solution and check your answer by substituting this approximate value in equation (2).

(C) Use Cardano's method to solve

$$x^3 - 6x^2 + 9x - 6 = 0 \tag{3}$$

Use a calculator to find a decimal approximation of your solution and check your answer by substituting this approximate value in equation (3).

(D) In step 2 of Cardano's method, show that u^3 is real if $\left(\frac{n}{2}\right)^2 \geq \left(\frac{-m}{3}\right)^3$.



› **Figure 2** Distance between two points.

› **THEOREM 1 Distance Formula**

The distance between $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ is

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

EXAMPLE 2

Using the Distance Formula

Find the distance between the points $(-3, 5)$ and $(-2, -8)$.*

SOLUTION Let $(x_1, y_1) = (-3, 5)$ and $(x_2, y_2) = (-2, -8)$. Then,

$$\begin{aligned} d &= \sqrt{[(-2) - (-3)]^2 + [(-8) - 5]^2} \\ &= \sqrt{(-2 + 3)^2 + (-8 - 5)^2} = \sqrt{1^2 + (-13)^2} = \sqrt{1 + 169} = \sqrt{170} \end{aligned}$$

Notice that if we choose $(x_1, y_1) = (-2, -8)$ and $(x_2, y_2) = (-3, 5)$, then

$$d = \sqrt{[(-3) - (-2)]^2 + [5 - (-8)]^2} = \sqrt{1 + 169} = \sqrt{170}$$

so it doesn't matter which point we designate as P_1 or P_2.

MATCHED PROBLEM 2 Find the distance between the points $(6, -3)$ and $(-7, -5)$.

› Midpoint of a Line Segment

The **midpoint** of a line segment is the point that is equidistant from each of the endpoints. A formula for finding the midpoint is given in Theorem 2. The proof is discussed in the exercises.

› **THEOREM 2 Midpoint Formula**

The midpoint of the line segment joining $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

The point M is the unique point satisfying

$$d(P_1, M) = d(M, P_2) = \frac{1}{2}d(P_1, P_2)$$

*We often speak of the point (a, b) when we are referring to the point with coordinates (a, b). This shorthand, though not technically accurate, causes little trouble, and we will continue the practice.